ActInf Textbook Group ~ Cohort 1 ~ Meeting 9 (Chapter 4 pt. 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 9 (Chapter 4 pt. 2) | 58:24

of these questions
i think we talked about the oh this is
chapter two
uh let's go to chapter four
okay
so we talked about belief policy state
i'm not sure if we talked about this
question
[Music]
in the discussion of active inference in
pomdp
belief updating about policies we find
that the posterior that minimizes the
free energy does it posterior at time t
oh we did discuss this
yeah we did discuss this last time
so
uh
this question
i think um
we did not discuss pi being a policy or
a model
last time or did we
i think we didn't get to this
yeah i don't think we discussed that all
right so we can open it up and um
we'll start here i guess with the the
most upvoted so the question reads
what is pi on page 69 the authors write
at each time step the current state is
conditionally dependent on the state at
the previous time and on the policy pie
currently being perceived pursued then
on page 71 they write thus we can
interpret the priors of equation 4.6

combined with the likelihood of equation
4.5 as expressing a model pi of a
behavioral sequence so which is it
policy or model
and then they suggest a rewrite of the
sentence they say that thus we can
interpret the priors of equation 4.6
combined with the likelihood of equation
4.5 and the transition probabilities b
sub tau pi
as expressing a model for a behavioral
sequence where the model is a function
of policy pi
and then there's some discourse here so
in this reframing can we say that the
model simulates the agent in the
environment as as if it had taken the
actions in the policy
um
and i don't know does anyone have any i
don't know who asked that question um
oh eric that was you that was your
question uh and then does anyone want to
maybe take a stab at answering that
question or we can continue reading what
was written here
uh so
what's written here
says that on page 69 the authors
also say policies here may be thought of
as indexing alternative trajectories or
sequences of actions that could be
followed on page 71 they defined the
likelihood in equation 4.5 as a matrix a
that expresses the probability of an
outcome they describe the priors of
equation 4.6 as the prior over the
initial state vector d

and beliefs about how the state at one
time transitions to the state at the
next time
matrix b and they also say that the
transitions are conditionally dependent
on the policy chosen because of this
conditional dependence we can see how
the policy influences the model and why
the authors use may use these the terms
interchangeably
so
does anyone have any
comments there
um i guess i would say that that um
answer in discourse
kind of um agrees with my that i
proposed which is
that strictly speaking
um we should treat the
pie as policy
but
it behaves as a model but once it's
executed it's implemented then
um
the model become the model has become a
function of the policy that's been we've
seen so far
so i i'd say that's a you know
that's compatible
um interpretation so i think strictly
speaking
um
the um
it would be better if the text was
consistent and called pi
a policy
and then um say yeah the model is a
function of path

that'd be my interpretation
yeah i definitely agree with that i
think that the model
is policy specific so so
yeah but they could do a better job of i
mean the variables
as we've seen are so
ambiguous anyway so it could definitely
be a lot better
uh and by the way uh about the topic of
consistency um actually i gave some
thoughts about the issue we were
discussing in yesterday's math learning
session
and i also consulted some other papers
and active inference and
in all of them the notation for matrices
and vectors
is used consistently as in almost every
linear algebra textbook so i'm seriously
beginning to suspect that every instance
of a matrix or a vector not written in
boldface is a typo either in chapters or
appendices because otherwise i really
cannot find any justification behind
using two different types of notation so
i'm going to use this assumption as my
prior for the rest of the book unless
someone has any belief updating
observation she wants to share
so just to summarize for for the people
that weren't there yesterday we had a
big debate about what does the bold
um typeface signify
and it wasn't so much a debate but but
just like a mutual trying to come to a
mutual understanding so it's this
question here does bold lettering mean

this is a matrix but seemingly also many
non-bold letters are matrix or does bold
lettering mean sufficient statistics of
um as as on these page examples
and it really was uh
yeah we couldn't come to a
single unified answer there
because there are many instances where
it seems like it should be a matrix but
it's not bold
and there are many instances where
um sufficient statistics like are used
but it's also not bold so
so yeah we couldn't really conclude that
um but i think ali that that's a good
assumption there
uh so we also talked about this
a little bit last time but we didn't
really get fully into this question what
is the use of categorical distributions
in equation 4.5 the second line is
supposed to explain the cat notation but
i have problems to understand the
advantage of the notation could someone
explain it in simple words
does anyone want to try to explain that
i can pull up the equation
uh well i think every regular matrix and
vector
is
by definition an array of ordered
elements
uh but in a categorical distribution uh
we don't necessarily assign any
any specific order to the elements so
uh the reason behind using the
categorical categorical distribution
might be

uh
to just jettison any
uh orderness in the elements and treat
them as just
uh an on an unordered set of the
probabilities
i wrote an answer to this one if you
want to pull that out
so you can just write that
so i hope this is helpful
it's basically
the way i see it is you have the math
the math notation there
but when you need to actually implement
that you need to carry it out then you
need to turn that into operations you
can perform
and so the um when you call when you say
that
a is this categorical
object that means that you you're
turning it into an actual
matrix
with elements and so the ordering is
important actually because
then the ordering of the elements in the
matrix says how that matrix is going to
apply to
um the objects it applies to which are
the um the belief state s and then the
your observation
o
so um
basically it says
the way you carry out
inferring what o will be
the probability of o given a belief
state is you do a matrix multiply

and
the because our in in this you know in
this example
um
the the um states are not continuous
they're categorical
then you need a categorical matrix you
know elements in order to do
map from the categories of
belief states to the categories of
observation states and those are all
those are both distributions
that gets transformed by the
matrix
thanks eric that was super
helpful
does anybody have any further question
or comment on this one
uh sorry eric but
isn't the categorical notation special
case of multinomial distribution
categorical distribution special cases
of multinomial distribution
and
i don't know that
yeah because that was my understanding
that the main difference between uh any
categorical distribution and uh
well
i mean uh the the lack of
for instance if we
of course you're right that in any
matrix
we necessarily have some ordered
elements in order uh for us to be able
to compute
the math on those matrices uh but on the
other hand uh

for categorical distributions
at least that was my understanding that
uh we don't necessarily
treat them as
ordered elements such as
i don't know the ordered
vectors or ordered matrices as
the normal
matrix because uh if that was the case
well
we could just uh designate this this
distribution as just a regular matrix
right so uh i don't get what's the
reason behind using
uh the notation here a
cat
stephen can you mute you're super loud
over there
sorry
uh
yeah i i guess so
um
i mean yeah so the mathematical notation
doesn't have an ordering it just says
distribution is related to another
distribution through this
um
this this relation and it's when you get
to the mechanics of how do you
how do you do that that's where you care
that you have the matrix with with
the vector ordering so yeah i don't know
what
cat
a actually means otherwise other than
it's just describing it it's saying
i don't think it's like an operator i
think it's just saying what it is

that makes sense
steven if you're trying to talk we can't
hear
i feel like steven is trying to share an
idea with us
but is unable to communicate effectively
through the technological affordances
he's been given at the time
like the gain on your mic is way up or
something uh so it's picking up more
background noise than you
so just looking at this equation
um
let's try that again
so just looking at this equation can you
guys hear me okay
yeah
yeah okay
so just looking at this equation
the second part
it's like the
it's like an ordered matrix a i j
um
where the probability of the
observations are
equal to index i and the states equal to
index j
something like that
um
i don't know some it looks like it is
ordered
um
yeah the matrix itself is ordered
yeah and so the second line is is
actually by how the authors are defining
this cat a
so so cat a represents this ordered
matrix a i j

um
yeah
that's at least what it what it says in
the textbook
so it says the likelihood expressing the
probability of observations conditioned
on time tau given states s
conditioned on time tau is equal to the
categorical distribution a
um
okay
uh
if i may
sorry sorry yaku
go ahead sorry my internet is a bit is
lagging a bit so there might be a delay
and i might break off but i just want to
comment that i think that the
cat notation is really just to
uh like
reinforce
the the notion that it's in discrete
time but i think it's
kind of
redundant in this case because
since we know that we are in discrete
time we could have just written
uh a bold to
uh signify that it's a matrix and when
it's a matrix it's probably not going to
be a continuous distribution anyway
um
so i think
cat
bolt a and bolt a is pretty much
interchangeable
so i think that some of the new math um
is using some of the affordances from

category theory more like some of the
the newer things and i think like the
forward
direction that
carl and thomas and some of the people
that really helped derive and
advance some of this math i think that
they're going to be leveraging category
theory methods like the renormalization
group and the pullback attractor um more
in the future and so that might be also
like
part of the reason that they are moving
toward this categorical notation so just
a little maybe foreshadowing there too
and may i just make some one additional
note about this categorical distribution
well the thing that confused me about uh
this this concept of categorical
distribution is uh the wikipedia's
definition of it because here it says
there is no innate underlying ordering
of these outcomes but numerical labels
are often attached for convenience in
describing
the distribution
so that was my confusion
um i don't think it would be the first
time that we've seen a traditional
mathematical notation slightly and
somewhat abused in active inference
probably also won't be the last
but definitely i think
it's a point that we can
put forward for to the authors for
additional clarification
um
if there's no further

anybody has anything else to say about
the categorical distributions here
if not we can move on to maybe the next
question
so this next question reads in message
passing is there a decay of information
as the distance between the variable x
and individual markov blanket
constituents increases
is implementation of information decay
in message passing an option for model
implementation
um i think that this is a super
interesting question and it's something
that i've thought about before a lot
uh
but but i think um
yeah let me open it up to anyone and see
if they want to address it before we can
get into some of the discourse
well doesn't that all
all depend upon the
the
chain of loss between one stage of the
message passing in another and if you
have no loss
then you don't have any decay
but you know the more stages you have
where you get a little bit of
diffusion or loss then you're going to
have decay
just a function of the parameters
well so our message is passed within
the same markov blanket or between
things that are partitioned with their
own markov blanket that's another
question
uh i also agree with eric

as i
mentioned
yesterday uh
i also believe that uh well
this message this information loss is
not accounted for at least inherently in
active inference formalism uh because
it depends on the specific situations
that
those information propagate because
otherwise uh
we couldn't have uh any uh
general or all-encompassing
formalism for accounting for these mess
information laws
so we see in this box
they talk about markov blankets and
the message passing so here it's
variational message passing this
involves messages from all constituents
of the markov blanket of x including the
parents via the conditional probability
of x given its parents and the children
the latter depends on the conditional
probability of the children of x given
all of their parents which include x
note the expectation includes the
children and parents of the children as
parents of the children x we divide by q
of x to ensure the expectation includes
the blanket only
so i think the messages are like all of
the things inside of the blanket send a
message to something else
in another blanket at least that's how i
interpret
um this
way of active inference and so

it would make sense that um
like from within the blanket does
everyone have the same is there like a
high degree of mutual information shared
in the markov blanket i think like
conditional dependence is what what um
you know qualifies as as
like conditional independence is what um
establishes the markup blanket so within
constituents under the blanket do they
all have all of the same information all
of the time that's something that's kind
of confusing to me
you know one thing i'd point out about
figure
4.3
so in the previous chapters
um you know we have this this picture of
there being the world and your model of
the world
and and that and those being
and your model of the world
is
a markov blanket itself and it
interfaces with the world through the
the um observation variables and the act
and the action variables
this picture doesn't have that breakdown
so it's got state there but it doesn't
say what is that is that state all
internal state is it external state or a
mixture of them and it's apparently it's
mixture because you've got policy that
can affect state which means that you're
going to have action is you know is
going to be part of the state transition
as well as and then you have the
observations coming out at the bottom

um but i think if we're going to talk
about markov blankets then i would like
to see this picture
be exploded into the internal you know
the state of the machine versus the
state of of the external world and how
those inter interact with one another
because that's what the markov blanket
does is it compartmentalizes the
information
well and and so that is the the pomdp uh
model and figure 4.3
um and this really figure 4.4 um
is like makes me scratch my head quite a
bit this is the image that is going to
um like represent
uh
message passing in a bayesian framework
and so um
the caption says on the right
dependencies between different variables
in the belief updating scheme outlined
in the main text
intuitively current beliefs about states
under each policy at each time are
compared with those that would be
predicted given beliefs about states at
other times this is one
and current outcomes to calculate
prediction errors these errors then
drive updating beliefs this is two
given beliefs about states under each
policy we can then calculate the
gradients of the expected free energy
this is three
um
these are combined with the outcomes
predicted under each policy omitted from

the figure to compute beliefs about
policies this is
for
and then using a bayesian model average
we can then compute posterior beliefs
about states averaged over policies this
is five
um
yeah
so
uh and this kind of leads us into the
next question but like i i i'm not able
to really interpret this figure 4.4 in
any better way so if anybody has like a
plain english
description of um how
this works i would love to hear it
we can also um we can talk a little bit
about um the a question that was raised
about this figure and also about this
like squiggle sigma
uh because there's a great like i don't
know i mean
we kind of had a hard time figuring out
the equation and the squiggle sigma and
what that actually stands for so we
could move to that
discussion because i think it's maybe
related to this message passing
discussion but let's read the discourse
here first
um on the
message passing question
so
in the discourse
what is the mechanism for signal decay
so this is um
information decay implementation of

information decay
is the message a thing behaving as an
active inference agent itself or a
special piece of information not
behaving as an active inference system
itself
is there a blanket impedance mismatch i
think brock contributed this i'm not
sure if he's here today
um and then message passing is
described through active inference
usually is a hierarchy not a lateral
transfer like within constituents in the
blanket so the key thing here is how you
define the markup blanket either there's
a petition partitioned between particles
cells or whatever or there's no
partition in blanket and everything
under the blanket is conditionally
dependent does conditional dependence
preclude message passing are message
passed within nodes in the same blanket
i have not seen this and would love if
someone pointed me to some references
so that's that was me i wrote the last
part um
because i haven't seen
message passing under the same markov
blanket it's always been like from one
markov blanket partitioned object to a
different markov blanket partitioned
object
but that moves us into um if anybody has
any comments feel free well i would say
message passing is used to do the
inference so it's within a markup
blanket i mean that's
that's pretty common i think in the

message in them
the way message passing is used
so here as it's shown in figure 4.4 like
although this is totally like beyond me
to describe
here this is a temporal message passing
so it's like from one time point to
another time point
um and then on the left it depicts the
hierarchical
um like a hierarchical expansion
collapsing over time steps but that a
higher level network might predict the
states and policies at the lower level
and use these to draw inferences about
the context in which these occur so the
way that i've always seen the message
passing used is from one time point to
another or from one layer of a hierarchy
to the next layer of a hierarchy and
i've not seen it passed within
like
constituents under the same markov
blanket so eric if you have references
that depict like some kind of message
passing in active inference that's not
through time steps or through
hierarchical levels i would love to see
that because i have not seen it
well i'll just look at go back to box
4.1
where they talk about the variational
message passing and how you
you get information and i would say
within a blanket you've got various
nodes
and they they have this hierarchical
parent child relations

and in order to do inference about hey
what's you know what's our belief in one
of these parents you got to kind of go
up and down and say well what are the
children of that and what do the other
parents think
so that's all within within a single
markov blanket the parents and children
the hierarchical relation and the
message passing happens to to circulate
that information within the blanket
that's how i i read
and i also put a link here to a paper by
champion at all
in which uh they have actually
explicitly
uh defined uh the active inference or
better say reformulated the active
inference formalism according to uh
variational message passing and uh if
you just give me a second i can find
uh exactly the place where
they have stated
this
in plain english
oh
you can continue with the other
questions if you like
but i need a moment to
check this paper sure yeah i'll check it
out also and yeah thank you for that so
like when i'm reading this variational
message passing this involves messages
from all constituents of the markov
blanket of x so the message is coming
from all the constituents in the blanket
but it doesn't say where the message is
going

so it does make sense like that i guess
it that it that it's an interchangeable
like or within would be a better word
than from like within all constituents
of the markov blanket because i was
reading it as like the message is coming
from everything under that blanket which
i guess is an incorrect interpretation
um so thank you eric for pointing that
out because i was like what
so this is um a really hard question
that i looked to try to answer um
actually a bunch of us looked yesterday
a lot so it says in equation 4.10
the sigma variable
um
squiggle sigma we'll call it whatever
i'm not sure how to say it
is used to describe the difference
between the natural log of observations
conditioned on policy and preferences
what is the function of this variable it
also comes up in figure 4.4 the message
passing figure we're just looking at and
it would be great to have a verbal
description of this figure what other
papers or equations use this squiggle
sigma and uh we looked at length like
through the entire textbook through um
the other chapters it refers to like
we'll unpack this later in chapter seven
we looked through chapter seven we
looked through the appendices we saw a
variable that almost looks like that but
i think it's a gamma it looks like a
little bit more fancy than this squiggle
sigma and could not find
that

at all so we went to unpack equation
410
um
and so
here is the
equation itself
um
i'm not sure how to put this up next to
the actually maybe i'll put up here
419.
so and this equation 4.10
is
a rewriting of equation 4.7 in linear
algebraic form
so
we tried to to kind of unpack this a
little bit so the first line
states the prior probability for each
policy this is that pi sub zero is equal
to the softmax function sigma times the
negative expected free energy g
um and so that's this first line the
next line
is the expected free energy conditioned
on policy g sub pi is equal to the
entropy or negative expected log
probability
h times the states conditioned on policy
and time which is s sub pi times tau or
both of those sub pi and tau
plus the observations conditioned on
policy and time this o sub pi times tau
times the beliefs conditioned on policy
and time
which is the squiggle sigma maybe sub pi
times tau so so we were kind of unsure
if beliefs is correct here we kind of
pulled that out of the um legend for

figure 4.4
but if anybody um knows and and just to
maybe unpack that a little bit more the
beliefs conditioned on policy and time
the squiggle sigma sub pi times tau is
equal to the difference between the
observations conditioned on policy and
time o sub pi times tau and preferences
conditioned on time c sub pi or sub tau
so we were unsure if belief is correct
here also
um
and so if anybody knows if beliefs is um
the difference between observations and
preferences that would be great to have
some kind of feedback or input here
because we couldn't really reach a
conclusion
well i just i guess one one
tiny step
uh toward figuring this out i would say
is that
um that funny squiggle thing there
has to do with the c
we've got uh we've got this
probability of observations given c and
c is this mysterious object that
expresses preferences which has also
been underexplained
that's so if you look at the two terms
of
the
expected free energy g
as a function of a function of policy
and 4.7 and then look at how it looks
like in 4.10 that sigma thing there
is
is it is expressed as is

we've got the dot which i guess is a
multiplier
by observation so that's the same as
that's like saying your probability of
observation given c so that's i guess a
major matrix multiplied version of
the um this log probability of um
sequence of observations given your
preference
preference
prior c
yeah so they do unpack it a little bit
more in equation
4.7 or it seems to be like maybe we
could extrapolate from 4.7 if this is
beliefs because it is this natural log
of observations minus the natural log of
preferences but but up in the top that
just the free energy is um the entropy
or the maybe the dot product of the
entropy times the states
plus the observations
times or the dot product of the
observations and
maybe beliefs here um so i've not seen
like i can't recall a mathematical
representation of belief ever using this
squiggle sigma before and i looked
through
um the recent ryan smith paper and i
looked even at like the message passing
paper by fristen
and i was not able to really get any
additional like
references to this squiggle sigma at all
so why do you say belief as opposed to
preferences
well so it says here that um

the squiggle sigma
is
equal to the difference between the
natural log of the observations
conditioned on policy and time
and the natural log of the preferences
at at a certain time
so the sigma is defined by the natural
log of the preferences but it's more
than just the preferences
does that make sense eric
yeah
but that's not same as belief right
that's that's observations versus
preferences
right so the difference like
rate from
expected deviation or how much risk is
being taken relative to
the policy that's applied so because i
i mean it's the dot product is between
the observation and
that's wiggle sigma right so which is in
turn
like
okay this is what i've observed this is
this is what i expected so it's like a
divergence between the two
so how much more risk am i taking and
how far is just pushing me away from my
cream from minimizing free energy
so maybe it's something like that
whereas belief is wrapped up in the s
because that's that's your model you
know of the world
so risk is maybe the how the the
squiggle sigma is defined but then if
you look into this figure 4.4 like

that's that's why we used belief
so here let's let's i'll pull it up
right now
so i'm not sure how easy this is to see
but um
in the before we get to number one here
we have like at the second on the right
hand side the second to lowest level
is um
states conditioned on
policy and time i think i'm going to
read it off this bigger screen oh yeah
states conditioned on policy and time at
different time steps so like the the
current time is in the middle
the forward time step is on the right
and the backward time step is on the
left
um and so that's what we start with
so
intuitively current beliefs about states
under each policy at each time are
compared with those that would be
predicted given beliefs about states at
other times
um
so
maybe beliefs is this s well so and
that's kind of what what eric was saying
um and we know because it's defined
earlier this epsilon here is
a prediction error right so so what we
come to after step one it says and
current outcomes to calculate prediction
errors that's what we arrive at after
number one
um the errors then drive updating in the
beliefs that's number two so here we go

from the current state at at the present
time to a prediction error that drives
the belief updating
at this forward time step after number
two
um and then it says given beliefs about
states under each policy
we can then calculate the gradients of
expected free energy
three so so what gives us this gradient
of expected expected free energy is it
this
s sub type so s sub tau plus one or is
that a tau what i think that's a tau
let's see
yeah
so is it a future state not conditioned
on policy or here we get to this
squiggle sigma so it's a squiggle sigma
conditioned on policy and time plus one
so it looks like a belief update there
which is why belief made sense
so is that a risk update is that a
possible
interpretation there in this message
passing figure
well they say it's a gradient
which means it's saying
how much do we have to change our policy
i think in order to get uh our
objectives
yeah gradient makes sense to me as well
because uh it's how much information
you've gained right
so if you actually look at the change in
the state with respect to
so i apply some
let's say force or like let's just take

a simple example like if you have an
actuator
that just the applies force
and
all it does is
move moves forward and has to follow a
trajectory
so
the error is so at each time step if
it's a deterministic system you expect
it to follow a straight line and it's
deviating up away so that's epsilon
so you have to update okay this is how
far up i am from the state
that i'm supposed to be so i need to
apply slightly less
force or maybe a
force in a different direction
and
that seems so
it's also how much i have to over
correct in the future depending on what
i've done
currently right it's not just
uh so
if if i've applied some force right now
i might deviate in the opposite
direction and i have to come back so
i have to figure out how much force to
apply so
that's why i said
risk or yeah gradient
you would have to take the
uh change with respect to
the state change with respect to the
error so yeah that would be one
interpretation of it right so it's a
gradient

of the state change with respect to what
you have done
so i also heard the term information
gain in there um yeah yeah i mean i i
mean i'm just generally saying so in
neural networks for example we would
have like some sort of loss function
and we would calculate the weight change
with respect to the change of the loss
function right so like i have a weight
vector and then
uh this particular example is giving me
a certain amount of gradient
and i need to minimize this gradient so
i need to change the weights in turn so
the gradient actually
interfaces between this loss function
and all these weights
so effectively the gradient passes on
information as to how much the weight
has to change in order to minimize that
loss function that's why i said
information
and also as a side note
i think i found
this
sigma in ryan smith's paper too uh and
it is defined as the expected prediction
error
so
the expected prediction error versus the
actual prediction error like and and
they distinguish it from epsilon in the
ryan smith paper because i i did look at
that but i didn't find it in there yeah
it's in question in equation 27
under the section outcome prediction
errors

it may be that the expected means it's
an expectation over the um
the q distribution which is your
distribution over
um
belief states
yeah that's great ali like we were um we
ali and i and um some others worked on
this yesterday
a lot and uh it's great that it kept you
up late at night clearly ellie so you
went digging around for
for some more information that's awesome
yeah great okay perfect well that's that
resolves that very well
um
and then let's get into maybe the next
question
um
all right
so in equation 4.16
the x has a dot over it changed through
time but not the y
data
so there's some um
discourse on this
or maybe we should go to the next
question actually because it's maybe
more related so
figure 4.6 uses an epsilon for
prediction error as described in
equation 4.21
is this predictive processing framing a
part of the active inference model
or is this presented for contrast to
illustrate the simile similarities and
differences between active inference and
predictive processing

so this is equation 4.1 and this is the
epsilon but i think we see it way before
equation 4.21 like it's even there in
equation 4.4
um it says this predictive coding schema
is part of the active inference model
illustrating the hierarchical structure
of predictions and beliefs the authors
say one way to think about this is as if
we had equipped a predictive coding
scheme with classical reflex arcs at the
lowest level of the hierarchy and they
give this reference
in this setting active inference is just
predictive coding plus reflex arcs
does anybody know what a reflex arc is
because i'm a little bit lost with
respect to that or have any comments on
this prediction yeah reflex reflex arc
is uh
well basically
the stimuli that doesn't go through the
sensory cortex
i thought it was like a mathematical
construct
maybe you think of it as a control
system
no thinking it's just
it's like you know you bang your knee
your foot goes
up yeah that makes sense and it says we
we minimize three energy free energy
through action and the only part of the
free energy that depends on action is
the lowest level of prediction error in
this hierarchical schema
action fulfills descending predictions
by minimizing the error between the

predicted and observed sensory data
any additional comments here about
predictive coding and active inference
or we can look at this equation 4.16 the
x has the dot over it change through
time but not the y data
why is this
um
and here the discourse says
the top equation depends on f x and v
which is a deterministic function
describing how a hidden state changes
over time
the bottom equation depends on g of x
and v which describes how data are
generated from a single hidden state the
tilde indicates change through time the
dot indicates the first order derivative
any uh
comments here on this equation or your
question
thank you
so maybe we can ask one more question or
maybe a couple more so here it says on
page 63 it says specifically the
bayesian brain helps us frame the
problems that an agent engaging in
active inference must solve broadly
these are the problem of inferring
states of the world perception and
inferring a course of action planning
other than perception and action
planning are there other tasks or
challenges that the brain or organisms
engage in
how would we know
there's no discourse here but let's open
it up and see what you guys think

um yeah um people uh wonder hey uh
what's on that star out there
that's neither
i don't know is that is that inferring
state of the world i guess
not planning
and it could just be a hallucination of
the
generative model you're not really
measuring anything you're just measuring
yourself over and over again
that could be one thing
okay so here there's another question
for each of the graphical models in
figure 4.2 what is an intuitive example
for each structure
one would be disease diagnosis right
so you have symptom
that is causative of
or predictive of some condition
i think that was the example that they
gave in the text right
or did we talk about this last week
maybe
yeah i actually don't get the question
because reach of them we have an example
at least an example in the text so
yeah that's true
and i think we're going to maybe see
more examples as we go through the book
but just in case if anyone wants to see
some more examples uh actually that
paper i put the link in the chat
realizing active inference and variation
message passing
contains some more examples for each of
these diagrams
this is a fun last question maybe

um
it says
this reiterates that active inference
uses two constructs variational free
energy and expected free energy which
are mathematically related but play
distinct and complementary roles in your
own words what are the definitions of
variable variational free energy and
expected free energy and what role do
they play individually or together in
active inference
uh my view of expected free energy was
uh given a certain action that might be
taken it might produce this
or reduce the free energy or something
or it talks about some future
uh reduction on the free energy or
something like that
variational free energy is more
objective
i don't know how to put it so it's like
it's like okay actual kinetic energy
versus perceived kinetic energy kind of
thing
so you're rolling down a hill
uh yeah
like a ball rolling down a hill we can
actually measure
the kinetic energy at the bottom of the
hill right
if you put the ball at the top
expected free energy would be like
using a model to calculate that and then
finding out it
actually
corresponds if you look at the model
at least that's my view

so i don't know and probably like i
think about this in a lot of different
ways but
i think about like variational free
energy i mean this is one way to think
about it is like um
maximizing like the the trade-off
between epistemic value and pragmatic
value like right now and expected for
free energy is like maximizing that
difference at some point in the future
so you can like plan ahead like it might
be cold out so i'm going to take a
jacket
or like it's cold right now i'm going to
put my jacket on
um
those are kind of like that's how i
think about the difference between those
two
so it's ten thanks everyone for coming i
hope i didn't um you know muddle this up
too much uh trying to be a substitute
teacher for daniel um and uh it's ten so
we're gonna have tools right now in this
room but if anybody wants to continue
discussing these ideas and gather you're
welcome to
migrate up to one of the different
spaces
yeah and i think we'll stop the
recording now alex if you haven't
already